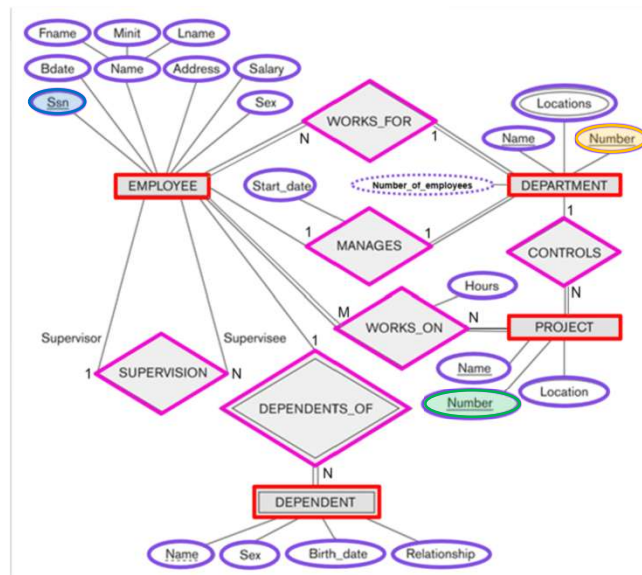


Chapter 4

Mapping an E-R Diagram to a Relational Model

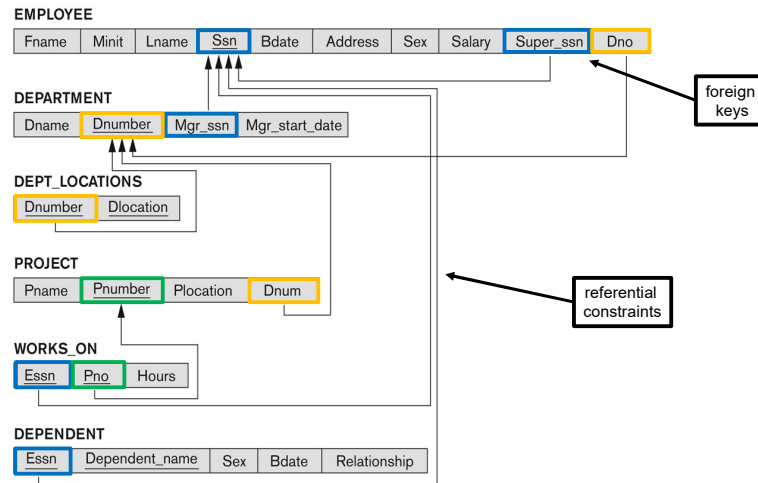
1

COMPANY E-R Diagram



2

COMPANY Relational Schema

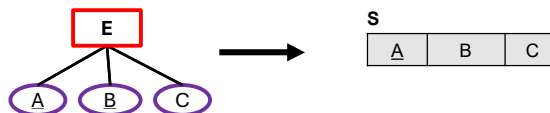


3

ER-to-Relational Mapping Algorithm

Step 1: Mapping Entities

- For each (strong) entity E in the ER diagram, create a relation S that includes all the simple attributes (or simple components of composite attributes) of E .
- Choose one of the key attributes of E as the primary key for S .
- If the primary key is composite, the set of attributes that form it will together form the primary key of S .

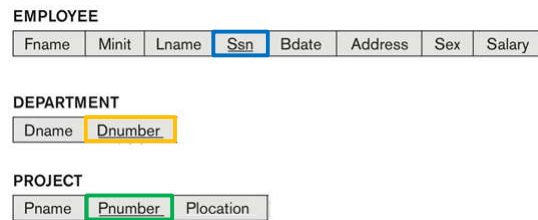


4

ER-to-Relational Mapping Algorithm

Example: Create the relations **EMPLOYEE**, **DEPARTMENT**, and **PROJECT** in the relational schema corresponding to the entities **EMPLOYEE**, **DEPARTMENT**, and **PROJECT** in the ER diagram.

- **Ssn**, **Dnumber**, and **Pnumber** are the primary keys for these relations

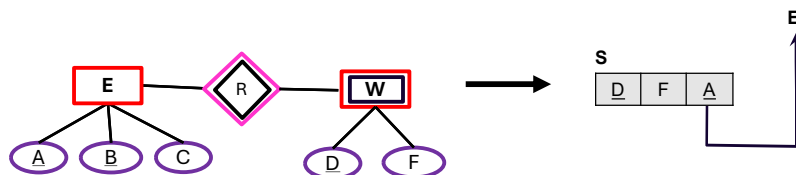


5

ER-to-Relational Mapping Algorithm

Step 2: Mapping Weak Entities

- For each weak entity *W* with identifying entity *E*, create a relation *S* and include all simple attributes (or simple components of composite attributes) of *W* as attributes of *S*.
- Include also the primary key attribute(s) of *E* as foreign key attributes of *S*.
- The primary key of *S* is the **combination** of the primary key(s) of *E* and the partial key of *W*, if any.



6

ER-to-Relational Mapping Algorithm

Example: Create the relation **DEPENDENT** to correspond to the weak entity **DEPENDENT**.

- Include the primary key **Ssn** of **EMPLOYEE** as a foreign key attribute of **DEPENDENT** (renamed to **Essn**).
- The primary key of **DEPENDENT** is the combination {**Essn**, **Dependent_name**} because **Dependent_name** is the partial key of **DEPENDENT**.



7

ER-to-Relational Mapping Algorithm

Step 3: Mapping a Binary 1:1 Relationship

For each binary 1:1 relationship *R* between the entities *S* and *T*:

There are three possible approaches:

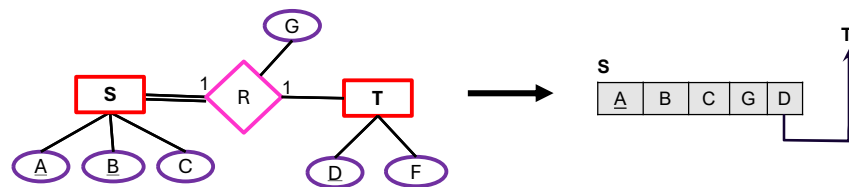
1. Foreign key (2 relations)
2. Merged relation (1 relation)
3. Cross-reference or relationship relation (3 relations)

8

ER-to-Relational Mapping Algorithm

Step 3: Mapping a Binary 1:1 Relationship

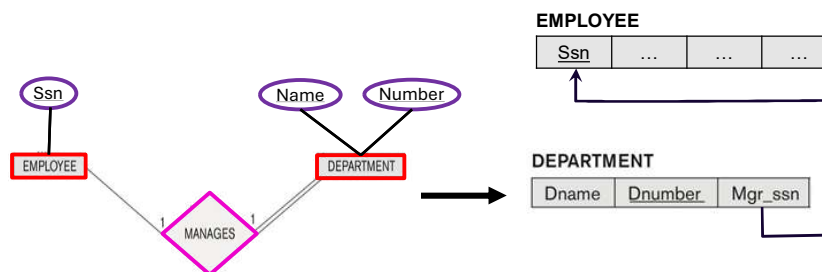
- Foreign key (2 relations) approach:** Choose one of the relations - say *S* - and include the primary key of *T* as a foreign key in *S*. It is better to choose as *S* an entity with total participation in *R*.
Include any attributes of *R* as attributes of *S*.



9

ER-to-Relational Mapping Algorithm

Example: The 1:1 relationship **MANAGES** is mapped by choosing the participating entity **DEPARTMENT** to serve in the role of *S*, because its participation in the **MANAGES** relationship is total.



10

ER-to-Relational Mapping Algorithm

Step 3: Mapping a Binary 1:1 Relationship

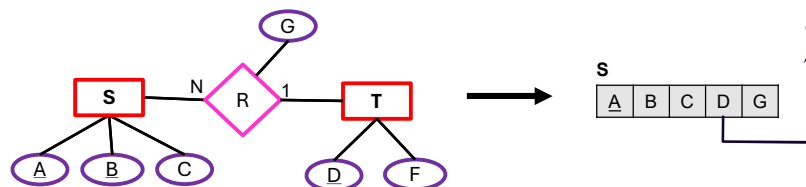
2. **Merged relation (1 relation) approach:** Merge the two entities and the relationship into a single relation. This may be appropriate when both participations are total.
3. **Cross-reference or relationship relation (3 relations) approach:** Create a third relation R for the purpose of cross-referencing the primary keys of the two relations S and T representing the entities.

11

ER-to-Relational Mapping Algorithm

Step 4: Mapping a Binary 1:N Relationship

- For each binary 1:N relationship R between the entities S and T where S is at the N-side of the relationship.
- Include the primary key of T as a foreign key in S .
- Include any attributes of R as attributes of S .

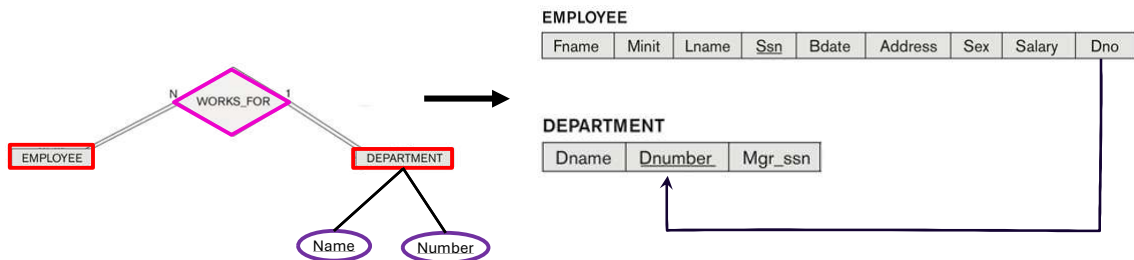


12

ER-to-Relational Mapping Algorithm

Example: 1:N relationships **WORKS_FOR**

Include the primary key **Dnumber** of **DEPARTMENT** as a foreign key in **EMPLOYEE** and call it **Dno**.

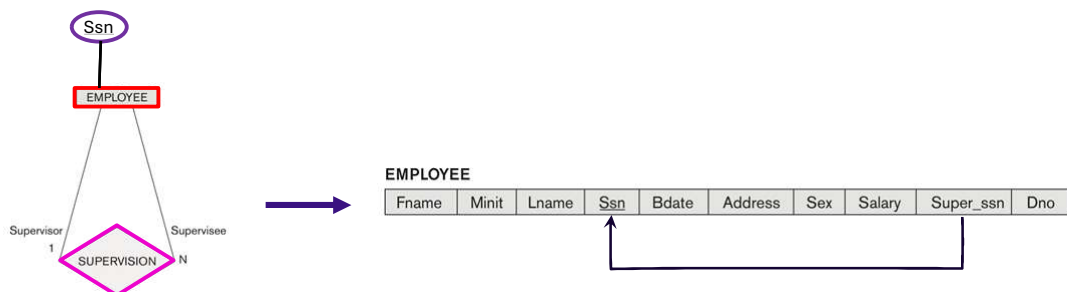


13

ER-to-Relational Mapping Algorithm

Example: 1:N relationship **SUPERVISION**

Include the primary key **Ssn** of **EMPLOYEE** as a foreign key in **EMPLOYEE** and call it **Super_ssn**.

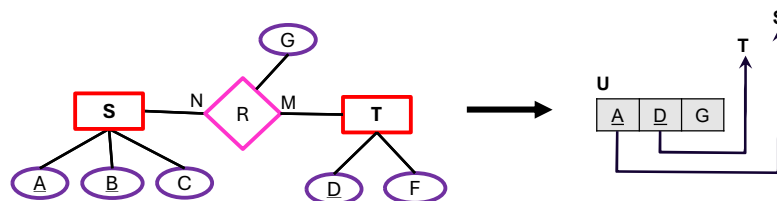


14

ER-to-Relational Mapping Algorithm

Step 5: Mapping a Binary M:N Relationship

- For each binary M:N relationship R between entities S and T , create a **new relation U** to represent R . This is a **relationship relation**.
- Include as foreign key attributes in U the primary keys of S and T ; **their combination will form the primary key of U** .
- Include any attributes of R as attributes of U .

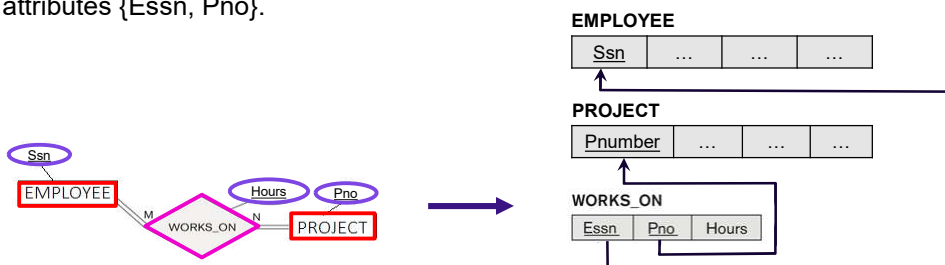


15

ER-to-Relational Mapping Algorithm

Example: The M:N relationship **WORKS_ON** is mapped by creating a relation **WORKS_ON**.

- The primary keys of **PROJECT** and **EMPLOYEE** are included as foreign keys in **WORKS_ON** and renamed Pno and Essn, respectively.
- The attribute **Hours** of **WORKS_ON** is included in **WORKS_ON**.
- The primary key of **WORKS_ON** is the combination of the foreign key attributes {Essn, Pno}.

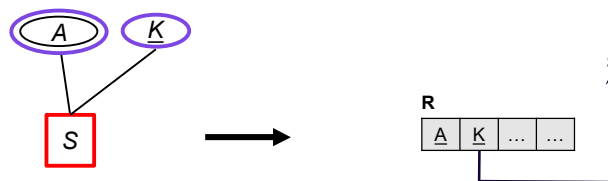


16

ER-to-Relational Mapping Algorithm

Step 6: Mapping Multivalued Attributes

- For each multivalued attribute A of entity S , create a **new relation** R .
- R will include an attribute corresponding to A and the primary key attribute K of S as a foreign key in R
- The primary key of R is the combination of A and K .

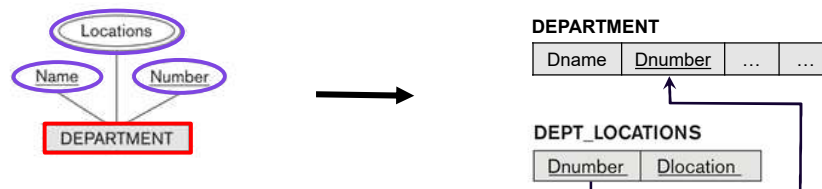


17

ER-to-Relational Mapping Algorithm

Example: The multivalued attribute **Locations** of **DEPARTMENT** is mapped by creating the new relation **DEPT_LOCATIONS**.

- The attribute **Dlocation** represents the multivalued attribute **Locations** of **DEPARTMENT**, while **Dnumber** – as a foreign key – represents the primary key **Number** of the **DEPARTMENT** relation.
- The primary key of **DEPT_LOCATIONS** is the combination {Dnumber, DLocation}.

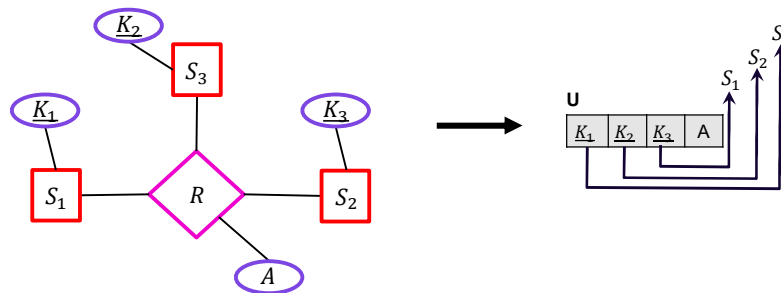


18

ER-to-Relational Mapping Algorithm

Step 7: Mapping N -ary Relationships

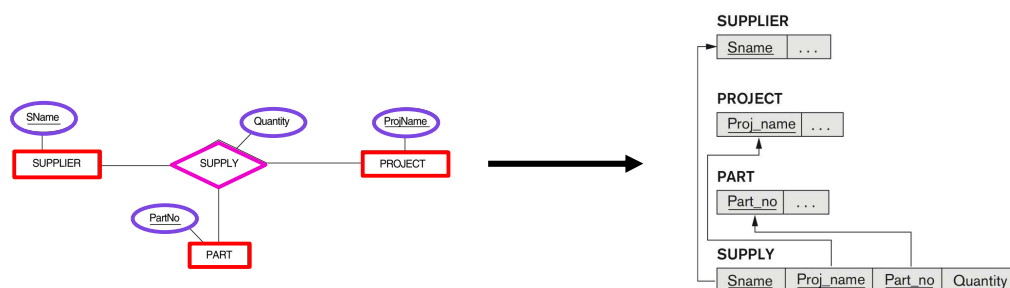
- For each n -ary relationship R between relations $S_1, \dots, S_n$, where $n > 2$, create a new relation U to represent R .
- Include as foreign key attributes in U the primary keys of $S_1, \dots, S_n$, whose combination will be the primary key for U .
- Include any simple attributes of R as attributes of U .



19

ER-to-Relational Mapping Algorithm

Example: The relationship **SUPPLY** can be mapped to the relation **SUPPLY** whose primary key is the combination of the three foreign keys {Sname, Part_no, Proj_name}

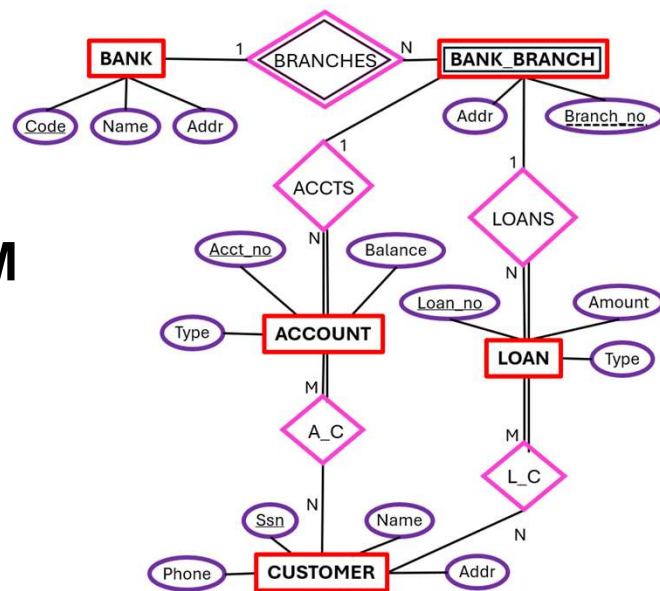


20

MAPPING EXAMPLE

21

BANKING SYSTEM



22

Q: List the strong (nonweak) entities in the ER diagram.

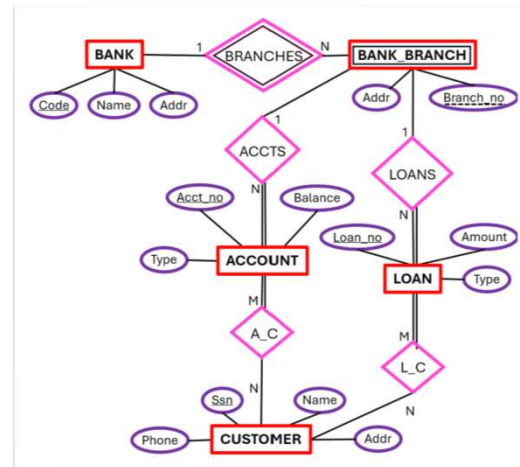
A: Entities: BANK, ACCOUNT, CUSTOMER, LOAN

Q: Is there a weak entity? If so, give its name, its partial key, and its identifying relationship. What does this mean ?

A: Weak entity: BANK_BRANCH. Partial key: Branch_no. Identifying relationship: BRANCHES. The combination of BANK Code and Branch_no together constitute a full identifier for a BANK_BRANCH.

Q: What is the relationship between the entities: BANK, BANK_BRANCH, ACCOUNT, LOAN ?

A: Each BANK can have multiple BANK_BRANCHES, and each BANK_BRANCH can have multiple ACCOUNTs and LOANs.



23

Q: List the user requirements that led to this ER diagram.

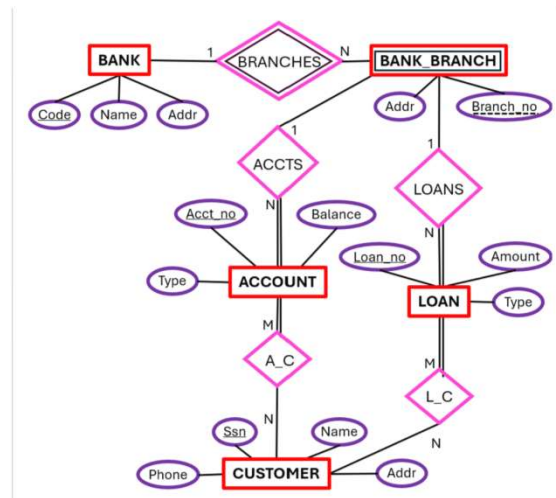
A: Each BANK has a unique Code, as well as a Name and Address. Each BANK is related to one or more BANK_BRANCHes, and the Branch_no is unique among each set of BANK_BRANCHes that are related to the same BANK.

Each BANK_BRANCH has an Address. Each BANK_BRANCH has zero or more LOANS and zero or more ACCOUNTs.

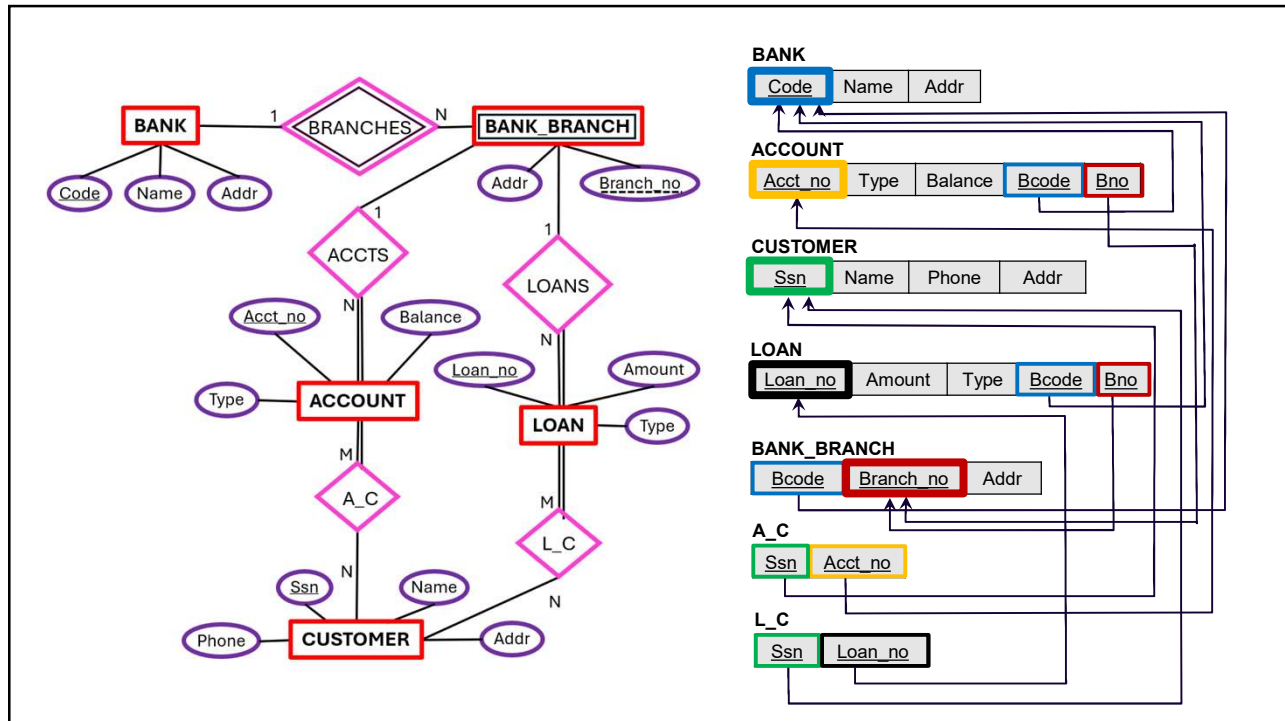
Each ACCOUNT has an AcctNo (unique), Balance, and Type and is related to exactly one BANK_BRANCH and to at least one CUSTOMER.

Each LOAN has a Loan_no (unique), Amount, and Type and is related to exactly one BANK_BRANCH and to at least one CUSTOMER.

Each CUSTOMER has an Ssn (unique), Name, Phone, and Address, and is related to zero or more ACCOUNTs and to zero or more LOANs.



24



25